OCN 682 / MBIO 612 • FALL 2026

AI-Assisted Community Data Sprint

OPIHI Intertidal Monitoring Data | Group assignment | 20% of course grade

The challenge

You are a small team of data-science consultants working with OPIHI (Our Project In Hawai‘i’s Intertidal), a long-term, teacher-supported intertidal monitoring program. Dr. Joanna Philippoff and collaborators want to understand what their historical surveys can tell them about ecological change, data quality, and sampling consistency. Your job is to deliver a transparent, reproducible exploratory workflow that the partners can reuse. Notably there are lots of datasheets in different formats taking from different people and you need to make it useable and reproducible.

Learning goals

Practice data importing, tidying, joining, summarizing, visualization, functions, Quarto, and GitHub. Use generative AI deliberately as a coding assistant in an ethical manner; independently test its suggestions, document errors and corrections, and respect community data stewardship.

Team and scope

Work in teams of 4. Each team selects ONE question below. Clean the datasheets, analyze a manageable subset of sites, years, or taxa as appropriate, and clearly explain the subset. You are not expected to analyze every record or answer all of Dr. Philippoff’s research questions.

Choose ONE project question

- A. Change through time: At one or a few repeatedly sampled sites, how did an observable metric (e.g., species richness, percent cover if recorded, or frequency of occurrence) vary across years? Plot some descriptive patterns.
- B. Sampling effort and data quality: How do the number of quadrats, transects, survey dates, or repeated surveys differ among sites and years? Which site-years are most complete? Visualize the data coverage.
- C. Species occurrence: For a selected taxon or group, where and when was it recorded? Make an occurrence or detection-frequency summary, accounting for sampling effort.
- D. Teacher/team effects: Does sampling completeness or consistency vary by teacher/team? Are some teams “better”? If so, how? Visualize information by different teams

At least one team needs to cover each of these topics. Sign up your name onto the signup sheet.

Requirements

1. Inspect and define

Download the OPIHI datasets; identify the observation unit, site, date/year, species or cover fields, and effort fields. Write out your specific question and define the metrics you are interested in.

2. Clean and document

Write R code in your Quarto document that imports the raw files (note, if files are in a proprietary format (like excel), convert as appropriate), cleans datasets as appropriate, and creates one clean "master" data file that will ultimately be used by OPIHI. Remember to keep raw data raw (i.e. don't delete anything manually in original datasets). However, there are LOTS of duplicate data across sheets. You need to find those and make sure no duplicates are not present in the Master sheet. The master sheet should be a finalized clean data sheet. Visualize a subset of the data in a beautiful tidy table. When cleaning the data check dates, data types, duplicates, missing values, taxon names (are there spelling errors that make it look like there are two species when there is really one?), and inconsistent site labels where relevant (do some use English names on some use Hawaiian names). Preserve raw data; explain exclusions. Make a data dictionary to go with the data set.

3. Make one reusable function

Create and use at least ONE short R function relevant to your workflow, such as a site-year effort checker, missingness summary, or taxon occurrence calculator as possible examples. Explain its inputs and output and test it. This will be usable for OPIHI in the future to check for errors in new incoming data.

4. Make figures and interpret

Produce TWO informative, labeled ggplot figures from your cleaned data addressing your ecological question. Include brief clear captions. This is just exploratory. Do not draw broad conclusions without statistics (statistics are not expected for this assignment).

5. Make it reproducible

Put the workflow in a GitHub repository on our OCN-682 organization page (only one repo needed per team) with appropriate relative paths, a README, a data dictionary/metadata sheet, and a short Quarto report. A reader should be able to rerun the work from start to finish using permitted data.

6. Audit the AI

Use an AI coding assistant for at least one substantive step (you can use AI for the entire assignment if you'd like). Test its output against the raw data and a manually checked small example; record the prompt, response or relevant excerpt, what was wrong or right, and your corrections.

Deliverables (one group submission + individual reflections)

- GitHub repository with: README; clearly organized R/Quarto files; one reusable function; cleaned analysis-ready output or code to generate it; concise metadata/data dictionary; and two figures. Make sure it renders correctly and that the html is published.

- Short Quarto report: question, clean dataset, cleaning decisions, two figures with captions.
- Group AI Developer Log: In either your README file or an additional .md file includes tool(s), selected prompts and outputs, one concrete AI mistake or questionable assumption (or a rigorous verification example), debugging steps, manual checks, and a disclosure statement (i.e. what models you used).
- Individual reflection (approximately one paragraphs per student): your individual contribution to the project, how AI helped or failed, what you personally verified, and when you would or would not trust AI-generated code.

Suggested milestones

- Oct 20: (launch): form groups, explore fields, select ONE question.
- Oct 27: cleaning script, initial effort checks, and draft function.
- Nov 3 (independent workday): finalize figures and Quarto report; test reproducibility.
- Nov 10 (before class): submit repository link and individual reflections; participate in class debrief.

Data stewardship and scientific guardrails

- Do not upload restricted raw data or teacher identifiers to public GitHub or an external AI system. Create Teacher IDs for your GitHub folder and a meta sheet to give to Joanna for which teachers belong to which IDs.
- Do not discard "odd" records without an explicit, reproducible rule and a record of what was excluded in the Quarto file.

Grading rubric (100 points)

Criterion	Points
Clear question, appropriate metric and scope	10
Data cleaning, documented decisions, metadata	25
Reproducible R workflow, function, GitHub and README	25
Two informative figures	15
AI Developer Log: evaluation, debugging and verification	15
Ethics, collaboration and individual reflection	10
TOTAL	100

Resources and submission

Relevant background that has used earlier versions of these data

La Valle et al. 2020 Hydrobiolgia , Ward et al. 2023 Plos One

Follow the syllabus AI-use policy: AI is required for this sprint but **not permitted for ordinary homework or the final independent project**.